EECE 571Q

Lecture 08: fault-tolerance and threshold conditions

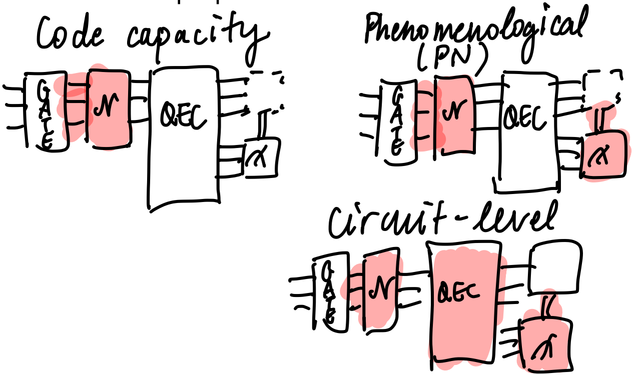
Thursday 4 June 2026

Announcements

- Quiz 4 Tuesday
- Final exam Monday 22 June 15:30-18:00
- Project topic proposals due next Friday 12 June

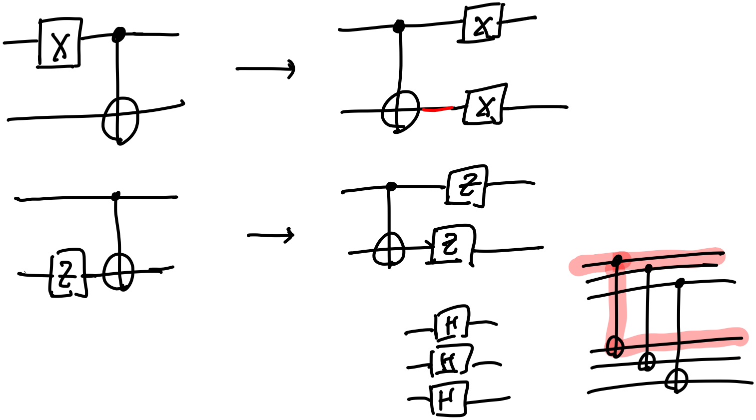
Last time

We defined three error models used to characterize and study fault-tolerant properties of codes:



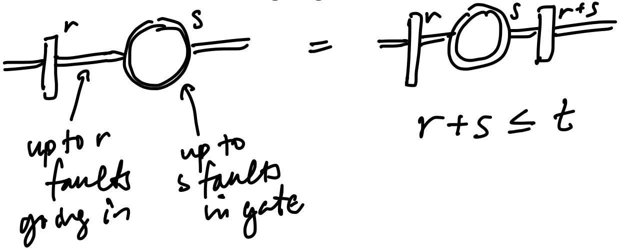
Last time

We propagated faults through various circuit elements



Last time

We defined what it means for circuit "gadgets" to be fault-tolerant.



similar criteria for 2-qubit gates, meas., etc.

Learning outcomes

1. Perform state preparation and measurement fault-tolerantly
2. State the threshold conditions of a code and concatenate codes to reduce failure probability
3. Carry out simulations in stim to estimate a code's (pseudo)threshold

Fault-tolerant state preparation and measurement

Is this circuit fault-tolerant?

$[[7, 1, 3]]$

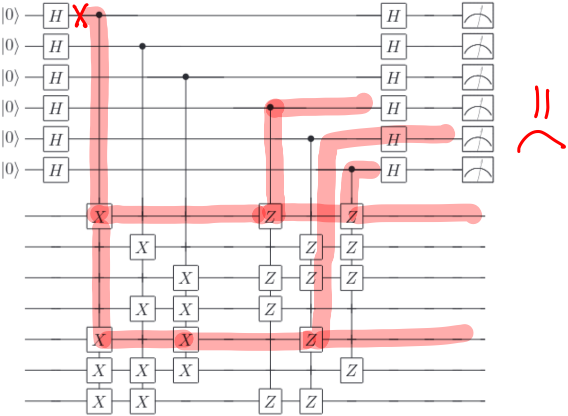


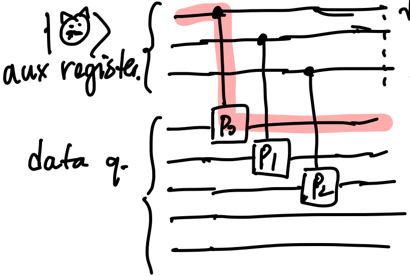
Figure 10.16. Quantum circuit for measuring the generators of the Steane code, to give the error syndrome. The top six qubits are the ancilla used for the measurement, and the bottom seven are the code qubits.

Screenshot: Nielsen & Chuang, chapter 10.5.8.

Fault-tolerant measurement

Idea: use cat states $\frac{1}{\sqrt{2}^n}\left(|0\rangle^{\otimes n} + |1\rangle^{\otimes n}\right) = |$ 🐱 $\rangle$

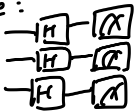
$P_0 \otimes P_1 \otimes P_2$



$\frac{1}{\sqrt{2}^n}\left(|0\rangle^{\otimes n} + |1\rangle^{\otimes n}\right)$ +1 eig. state (of $P_0 \otimes P_1 \otimes P_2$)

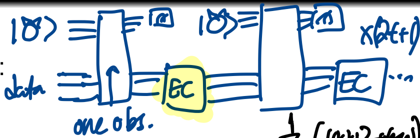
$\frac{1}{\sqrt{2}^n}\left(|0\rangle^{\otimes n} - |1\rangle^{\otimes n}\right)$ −1 eig. state.

measure:



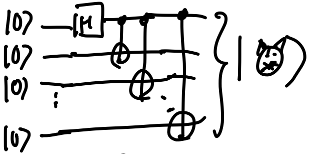
check parity.

Fault-tolerant measurement



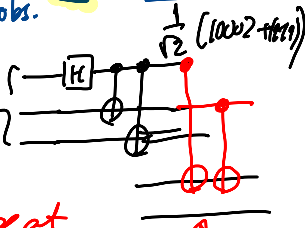
Solves one problem, but creates others:

- How to prepare the cat state?
- What if there is a measurement error?



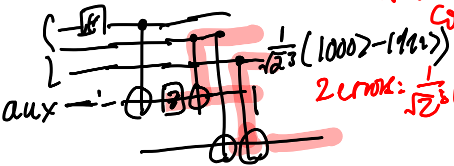
not FT

test if correct →



Solution: repeat until cat state is correct.

also not FT



2 errors: $\frac{1}{\sqrt{2}}(|000\rangle+|111\rangle)$

☆ many cat states

☆ many repetitions

Bit flip on aux: data qubit ok

Fault-tolerant syndrome measurement: Shor EC

What if we need to measure g stabilizer generators?

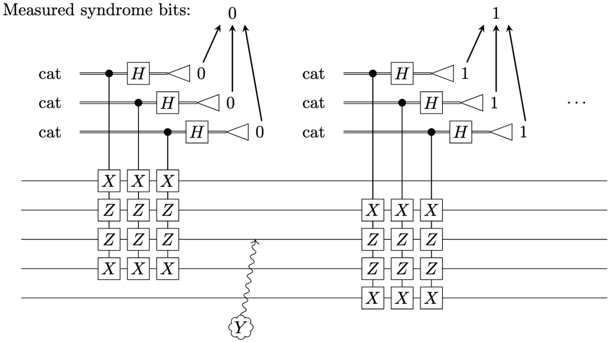


Figure 12.6: Repeating measurement of syndrome bits one-by-one for the five-qubit code. Because Y_3 occurs only after all repetitions of the first syndrome bit measurement are complete, the observed syndrome is 0110, which corresponds instead to X_4, instead of the correct syndrome 1110.

Image: screenshot from Gottesman Ch. 12

Fault-tolerant syndrome measurement: Shor EC

What if we need to measure g stabilizer generators?

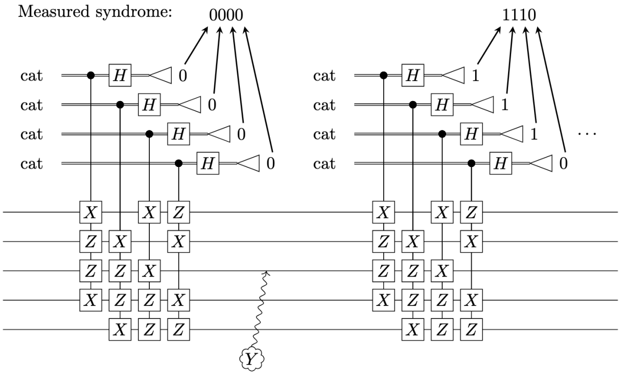
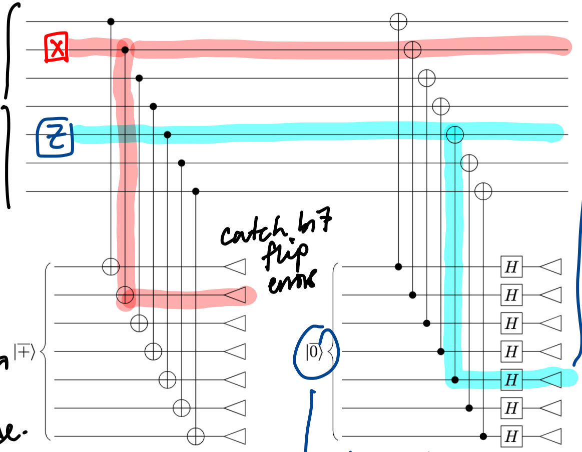


Figure 12.7: Measuring the full syndrome and then repeating for the five-qubit code. When the Y_3 error occurs after the first syndrome measurement, the second and future syndrome measurements give the correct syndrome 1110.

Image: screenshot from Gottesman Ch. 12

Fault-tolerant syndrome measurement for CSS codes

Special case: Steane error correction for CSS codes.



data (encoded in Steane code)

$\overline{CNOT}(|\overline{\psi}\rangle|\overline{+}\rangle) = |\overline{\psi}\rangle|\overline{+}\rangle$

logical states of Steane code.

catch bit flip error

catch phase flip errors.

apply Pauli correction to recover.

how to prepare FT?

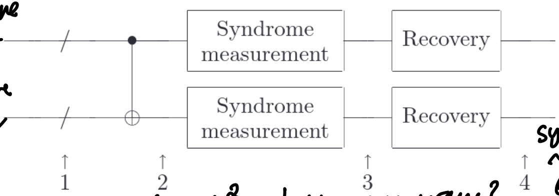
Image: screenshot from Gottesman Ch. 12

Thresholds

If one error occurs before/within this circuit with prob. p, what is the probability of *two or more* errors in the encoded first qubit at the output?

Steane code

Steane code



$\overbrace{6\times 10}^{\text{syndrome meas.}} + \overbrace{7}^{\text{recovery}}$

$C_0 \approx 70$

pre-existing error: $(C_0 p)^2$ how many ways?

two failures in logical CNOT: $C_2 p^2$ ways for two failures $7\times 7 = 49$

Screenshot: Nielsen & Chuang, chapter 10.6.1 (Fig. 10.21)

Thresholds

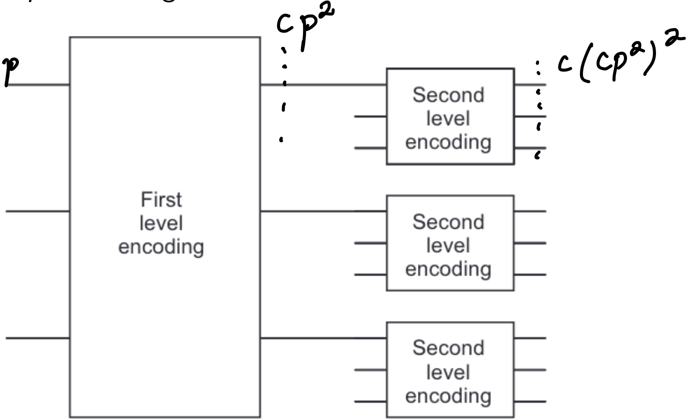
Can work through example of Nielsen & Chuang; key point is that probability of two or more errors in first encoded block is

$$\approx 10^4 \cdot p^2$$

As long as $p < 10^{-4}$, this is less than original error probability (more robust estimates closer to $\sim 10^{-5} - 10^{-6}$)

Concatenation

What if our p is too large? Concatenate codes...



Screenshot: Nielsen & Chuang, chapter 10.6.1 (Fig. 10.22)

Concatenation

Q: if you concatenate k times, what is the failure probability at the k'th level?

$$\frac{(cp)^{2^k}}{c}$$

Q: suppose g is the maximum number of operations in a fault-tolerant procedure. What is the simulation overhead?

$$g^k$$

Q: given a circuit with polynomial $p(n)$ gates, how many times do we need to concatenate to get total accuracy ϵ?

$$\frac{(cp)^{2^k}}{c} \leq \frac{\varepsilon}{p(n)}$$

Threshold conditions

$$\frac{(cp)^{2^k}}{c} \leq \frac{\varepsilon}{p(n)}$$

Can find k as long as $p < p_{th} = 1/c$ (*threshold condition*).

The size of the simulation circuit then becomes

$$O\left(\text{poly}\left(\log(p(n)\,\varepsilon)\; p(n)\right)\right)$$

Next time

Next class:

- Quiz 4
- The surface code

Action items:

1. Work on project proposal
2. Watch PrairieLearn for exam information

Recommended reading:

- For this class: Gottesman Ch. 12,13; N&C 10.6
- For next class: surface code blog post